Exhaustive three-model analysis tests performance complementarity

Every three-model combination is evaluated by equal-probability soft voting and ranked with the stored cross-dataset evidence without model-specific bonuses.

18	816	3	1
eligible models	three-model combinations	benchmark datasets	recommended trio

Agent contracts and responsibilities

versioned Markdown or control files

ROLE	DEFINITION FILE	RESPONSIBILITY
Chief handoff	chief_agent.md	Provides the accepted ranking evidence
Research advisor	research_advisor.md	Interprets stability and complementary errors
Combination engine	ensemble_top3_selector.py	Scores every unique three-model combination

Execution sequence

ordered file handoffs

01 **Align** → 02 **Enumerate** → 03 **Vote** → 04 **Score** → 05 **Recommend**

01	02	03	04	05
Align	**Enumerate**	**Vote**	**Score**	**Recommend**
three datasets	all 3-model sets	equal probability mean	stored 50-round evidence	external validation trio

Persisted evidence and stage handoff

project-state artifacts

ARTIFACT	ROLE	AUDIT / HANDOFF MEANING
ensemble_top3_selector.py	selection engine	exhaustive equal-probability soft voting
ensemble_top3_combination_ranking.csv	full ranking	816 evaluated combinations
ensemble_top3_selection.json	selection record	method, datasets, models, ranking and caveat
recommended_models	current trio	C_AMPs-predict / esm-AxP-GDL / pepnet_standard
amp_research_advisor.py	interpretation	performance, stability and future validation plan

Scientific boundary

The recommended trio is exploratory because selection used the same benchmark datasets; a formal performance claim requires an untouched external validation set.